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Multimodal sentiment analysis method based on image-text quantum transformer

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ABSTRACT

Multimodal sentiment analysis aims to recognize and interpret the diverse emotional information contained in different modal data. Although multimodal sentiment analysis methods have achieved significant results, they still have certain limitations in capturing the complex features of high-dimensional data. This paper proposes a multimodal sentiment analysis method utilizing the Image and Text Quantum Transformer (ITQT-MSA), which innovatively combines quantum computing and deep learning technologies to achieve more accurate emotion recognition and analysis. Firstly, a text-oriented parametric quantum circuit is designed that exploits quantum superposition and entanglement properties and combines the transformer model to achieve deep feature extraction from text data. Secondly, another image-oriented parametric quantum circuit is constructed and combined with the Visual Transformer (VIT) to adequately extract the emotional information in the image. Finally, effective alignment and the integration of features from both text and images are achieved by the designed cross-attention fusion mechanism. The experiments are performed on classical computers by designing quantum circuits within a simulated noisy environment, and the results show that the proposed method outperforms SOTA models.

1. Introduction

In recent years, sentiment analysis has become essential for interpreting human emotions and opinions. Traditional sentiment analysis methods focus primarily on unimodal data, which limits their ability to fully capture the complexity of human emotional expression. However, the emergence of multimodal sentiment analysis has enabled researchers to use multiple data sources to achieve more accurate and nuanced emotion detection.

For multimodal data, the synergy between text and images improves the model's ability to accurately assess sentiment. As shown in Fig. 1, relying solely on text or image modalities to detect sentiment may not capture the true intent behind a tweet. For example, in Figure (c), when considering the text modality, it appears to be a neutral statement. However, the man's smile in the image suggests that the sentiment of the tweet is positive. In Figure (b), when considering the image modality, the image does not convey any emotion, but the analysis of the text shows that it is a negative statement. In Figure (a), both the text and image modalities lack emotional information, indicating that the tweet is neutral. Relying solely on text or image modalities to judge sentiment is insufficient, so it is crucial to effectively capture the interactions between multimodal information. Multimodal interaction refers to the process of handling and integrating data from various modalities to effectively extract and utilize this information [1]. It mainly includes two components: intra-modal interaction and inter-modal interaction. Intra-modal interaction involves feature extraction and information processing within a single modality. Each modality has its unique features and structure. For example, in the text modality, common feature extraction methods include word embeddings (such as Word2Vec, GloVe) [2], sentence embeddings (such as BERT, GPT, T5) [3], and attention-based transformer models [4]. These methods can capture the semantics and syntax of the text to extract effective features. In the image modality, Convolutional Neural Networks (CNN) [5] are the most widely used tools for feature extraction. CNNs extract features such as edges, textures, and shapes, ultimately forming high-level image representations. Vision Transformers (VIT) [6] are also popular for image feature extraction, applying the original transformer architecture to image classification tasks based entirely on self-attention mechanisms. Inter-modal interaction involves the fusion and interaction of information between different modalities. The main goal is to effectively

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combine information from multiple modalities to improve system performance and accuracy.

As technology advances, the potential of quantum computing to process multimodal data has also attracted considerable attention. Quantum computing is characterized by its potential to solve complex computational problems. Unlike classical computers, which use a binary system based on bits, quantum computers use qubits, which can represent superpositions of both 0 and 1. This allows quantum computers to have significant parallel processing capabilities for certain types of problems. In addition, quantum entanglement is an important phenomenon in quantum computing. It occurs when multiple qubits are closely linked, and their states depend on each other. This enables fast information transfer and processing in complex computational tasks. Quantum interference uses the interference effect of phase information to increase the probability of correctly computed paths and eliminate erroneous paths. These quantum properties give quantum computers exponential speed advantages in tasks such as large number factorization, search problems, and optimization problems. Despite the presence of noise and errors, they have shown quantum advantages over classical computing in some well-designed tasks. In NLP, quantum-inspired language models are beginning to be applied to tasks like text classification and emotion analysis. The proposal of quantum self-attention neural networks (QSANN) [7] incorporates self-attention mechanisms into quantum neural networks. Using the quantum self-attention mechanism of Gaussian projection, QSANN demonstrates efficient and scalable capabilities on large datasets [8]. Compared to traditional syntax-based QNLP methods, QSANN can achieve better performance on larger datasets and is easier to implement on existing quantum devices. Quantum-inspired methods have recently been introduced into multimodal sentiment analysis, providing a new approach to model the complex interactions between different data modalities. Quantum theory provides a solid mathematical framework for representing and analyzing these interactions, making the fusion of multimodal features more effective. For example, a quantum-inspired multimodal fusion framework shows promising results in video sentiment analysis by using principles such as superposition and entanglement to model interactions within and across modalities [9]. These methods address the inherent semantic gap and complex association issues that traditional methods struggle to capture.

Despite the progress made, the field of multimodal sentiment analysis still faces several challenges. The first challenge is the "black-box" problem of neural networks [10], where the millions or even billions of parameters in a neural network are difficult to intuitively understand or interpret in terms of their specific roles within the model. Secondly, existing multimodal sentiment analysis methods suffer from insufficient expression of emotional features, as classical neural networks are currently unable to effectively capture the complexity of emotional information within multimodal data. The final challenge lies in designing and adopting suitable parameterized quantum circuits for feature extraction different data modalities. Given the significant differences in characteristics between modalities (such as text and images), the structure of the parameterized quantum circuits may have a critical impact on the effectiveness of feature extraction for specific modalities.

With regard to the first and second challenges, the advantages of quantum-inspired models mainly lie in the following two aspects. First, the Hilbert space constructed based on quantum theory allows emotional information (such as words, sentences, and images expressing emotions) to be mapped onto a complex vector space [10]. In this complex vector space, the model can leverage quantum superposition to represent multiple states of emotional information, thereby capturing the inherent ambiguity and vagueness in emotional data. Furthermore, the quantum entanglement property creates global correlations between features, enabling the parameters in the quantum circuit to carry global emotional information of the data. Second, the modules in quantum-inspired models are assigned physical meanings, such as superposition, entanglement, interference, collapse, and other quantum properties, providing the conditions for introducing interpretable parameters.

To address the final challenge, this paper designs different parameterized quantum circuit structures to fully accommodate the specific characteristics of text and image data, enabling deeper feature extraction from these modalities. In general, the contributions can be outlined as follows.

(1) To extract deep features from multimodal data, this paper proposes a parameterized quantum circuit for text and images. The circuit is combined with a multi-head self-attention mechanism. This combination enables deep feature extraction from both text and images.
(2) To better align and fuse text and image features, this paper uses text-image alignment and cross-attention fusion modules that can capture the fine-grained correspondences between images and text.
(3) Comprehensive evaluations and comparisons on public datasets demonstrate the advantages of the proposed approach. In addition, ablation studies were performed to confirm the effectiveness of the method.

This paper is organized as follows. Related work in sentiment analysis is summarized in Section 2. A full explanation of the proposed



Fig. 1. Examples of image-text pairs selected from tweets.

methodology is provided in Section 3. Experimental results, including dataset details and hyperparameter analysis, are presented in Section 4. Section 5 analyzes the model performance, including ablation studies, case analysis, and visualizations. Finally, Section 6 provides the conclusion of the paper and discusses directions for future research.

2. Related work

2.1. Classical sentiment analysis

Emotion recognition is significant in natural language processing and computer vision, with applications in social media analysis, customer feedback analysis, and intelligent assistants. Multimodal emotion recognition is the combination of different sources of information, such as text and images, to improve classification accuracy and robustness. Recently, significant progress has been made in this field through deep learning techniques, including multimodal fusion methods (early, late, and intermediate fusion) and deep learning approaches. Zadeh et al. [11] proposed a multimodal emotion recognition model based on CNN and LSTM, significantly improving performance by combining text and image information [11]. Zadeh et al. [12] introduced the Memory Fusion Network (MFN), effectively handling multimodal features using intermediate fusion to improve accuracy [12]. Tsai et al. [13] propose a multimodal transformer model, which takes advantage of cross-attention fusion mechanisms for the deep fusion of text and image features., further enhancing emotion recognition [13]. Yu et al. (2019) introduced an aspect-sensitive attention and fusion network, effectively modeling intra-modal and inter-modal interactions, including aspect-text and aspect-image alignment [14]. [15] developed an attention memory network for interactive learning of text and image features, integrating them through a multimodal feature fusion module with a multilayer perceptron and stack-pooling. Yang et al. (2021) used an emotion-aware multichannel graph neural network based on the global characteristics of the dataset for multimodal emotion analysis [16].[17] proposed a multimodal self-supervised learning-based emotion recognition model, significantly enhancing performance in multimodal emotion recognition tasks. Nguyen et al. [18] introduced a multimodal contrastive learning-based emotion recognition method, utilizing contrastive learning strategies to extract more contextual information from multimodal data, improving accuracy and robustness [18].

2.2. Quantum sentiment analysis

In the area of sentiment analysis, hybrid quantum-classical machine learning methods have shown unique advantages. Recently, researchers have explored various quantum computing techniques to enhance sentiment classification performance. Quantum theory shows unique advantages in modeling dialog and multimodal interactions. For example, [19] introduced the Quantum-Inspired Interaction Network (QIN), which integrates the mathematical principles of quantum theory with the LSTM architecture to model both intra-sentence and inter-sentence interactions. [20] designed a novel model called DMATT, combining it with Bi-GRU to form the Quantum-Inspired DMATT-BiGRU model, introducing static and dynamic density matrices to capture interactions and feed them into GRU and attention layers, respectively. Inspired by quantum entanglement, [21] introduced the Entanglement-Driven Fusion Neural Network (EFNN), a quantum probabilistic neural model. [22] developed a Quantum-Inspired Multimodal Fusion (QMF) model, formalizing intra-modal word interactions as quantum superposition and cross-attention interactions as quantum entanglement at different stages. Based on the QIN model,[23], [24] introduced the Quantum Analogue Multimodal Network (QMN) framework, utilizing various quantum probability forms and LSTM structure. Based on the CFN model [25], [26] proposed a QPM framework designed for the joint classification of multimodal sentiment and sarcasm. Inter-modal interactions are treated as quantum interference, and judgments of sentiment and sarcasm are seen as incompatible measurements. The Quantum Fuzzy Neural Network (QFNN)[27] combines Classical and Quantum Neural Networks (QNN) with fuzzy logic. It captures sentiment and sarcasm features using complex numbers in the Fuzzifier and utilizes QNN in the Defuzzifier for prediction.

Despite these achievements, classical sentiment analysis methods and quantum-inspired methods still face limitations in capturing high-dimensional complex features of different modalities. To address this, parameterized quantum circuits are proposed. They are applied to both text and image modalities. The quantum circuits effectively capture contextual relationships in text sentiment information and image sentiment features. This further improves the performance of the sentiment analysis model.

3. Methodology

3.1. Overview

This paper proposes a bimodal (text and image) sentiment recognition model, as shown in Fig. 2, combining deep learning and quantum computing techniques. The text learning module uses the output from the penultimate layer of the pre-trained T5 model together with a quantum transformer. For a detailed explanation, please refer to Section 3.2. The images learning module extracts features through the ResNet50 network and converts them into sequence form for further processing by the quantum transformer. For related details, please refer to Section 3.3. To address the problem of cross-modal semantic association caused by feature distribution differences between text and visual modalities, this paper uses a text-image alignment module that aims to optimize unimodal representations prior to feature fusion, thereby improving the consistency of modal representations.For specific details, please consult Section 3.4. On this basis, the model further introduces a cross attention feature fusion module, which maintains the original feature hierarchy while achieving deep integration between different modalities. By integrating and analyzing sentiment information from text and images, this model achieves efficient and accurate sentiment recognition.For further details, please refer to Section 3.5.

3.2. Text feature extraction based on quantum transformer

The quantum text learning module is a core component of the bimodal sentiment recognition model, primarily responsible for processing and analyzing text data to extract sentiment-related features. This paper proposes a text oriented parametric quantum circuit structure that, combined with the transformer, processes features extracted by the pre-trained T5 model to increase the model's depth of text understanding and feature representation capabilities.

The T5 [28] model, a highly efficient pre-trained language processing model, is used for initial text feature extraction. First, the raw text is tokenized and converted into an input sequence X. Then, T5 converts X into contextual representation $H^{(L)} \in R^{n\times d}$through embedding layers and continuous L-layer transformers, where n denotes the maximum sequence length and d represents the hidden layer dimension.

$$T = SentencePiece(X) \tag{1}$$

$$E = Embedding(T) \tag{2}$$

$$H^{(0)} = E + positionalEncoding(E) \tag{3}$$

$$H^{(i)} = Transformer\left(H^{(i-1)}\right),\ i \in \{1, \ldots, L\} \tag{4}$$

The features extracted by the T5 model will be further processed and enhanced by the quantum transformer, leveraging quantum computing to enhance feature representation and sentiment analysis capabilities.

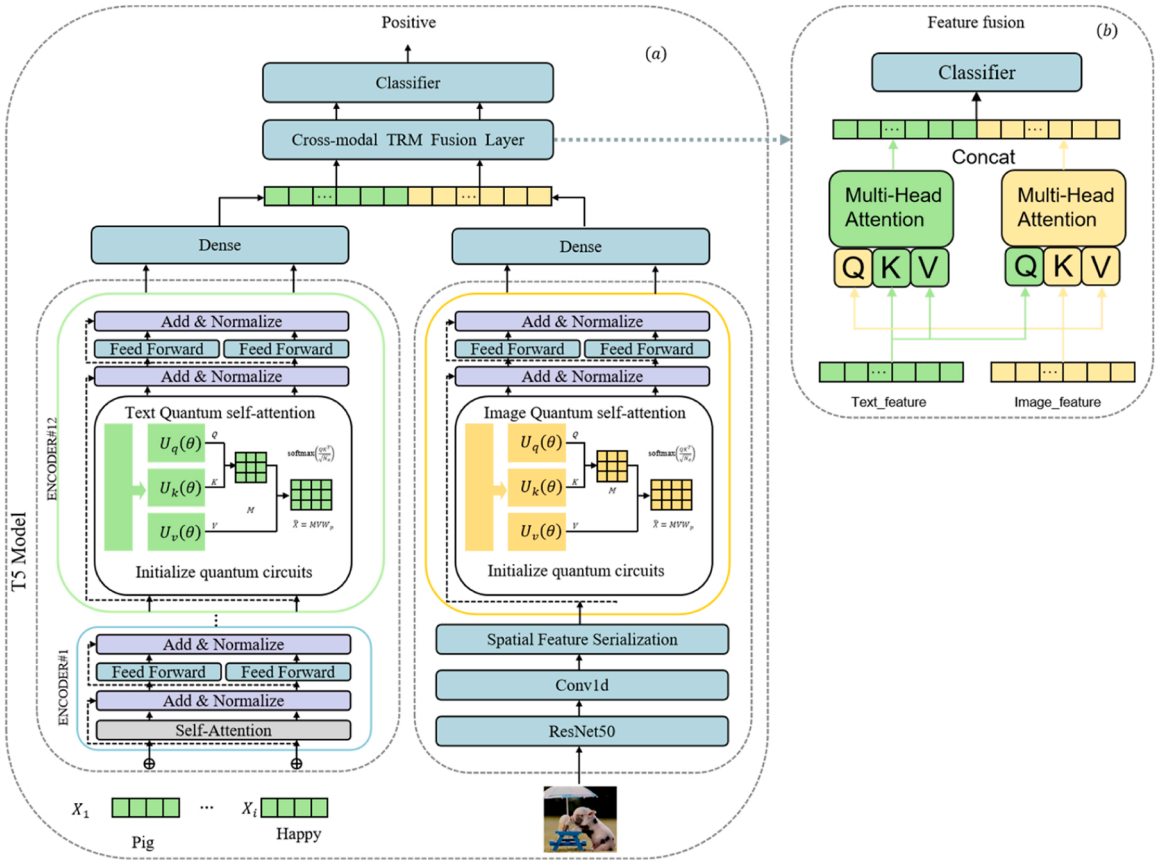


Fig. 2. The framework of the proposed ITQT-MSA model.

As shown in Fig. 3, it optimizes the traditional transformer's multi-head attention mechanism by incorporating quantum computing techniques. In the classical multi-head attention mechanism, three separate weight matrices are applied to transform the input into q, k, and v matrices. However, within the quantum multi-head attention mechanism, the three weight matrices are replaced with parameterized quantum circuits U_q,U_k,U_v. First, classical data is encoded into quantum states. Specifically, for classical data $H^{(L)} = \{h_1, \cdots, h_n\}$, $h_j \in R^d$,$1 \le j \le n$. For a certain sample h_j, the angle encoding method [29] is used to map the classical data to the rotation angle θ_j of the qubits.

$$\theta_j = \pi \times h_j \tag{5}$$

each qubit is encoded using a *Pauli-X* rotation gate $R_x(\theta)$, defined as:

$$R_x(\theta_j) = e^{-i\theta_j X/2} = \cos\left(\frac{\theta_j}{2}\right) I - i\sin\left(\frac{\theta_j}{2}\right) X \tag{6}$$

where i represents the imaginary unit, X is the *Pauli-X* matrix, mathematically represented as $X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, θ_j is the rotation angle, and I is the identity matrix. Therefore,the encoding operation is expressed as:

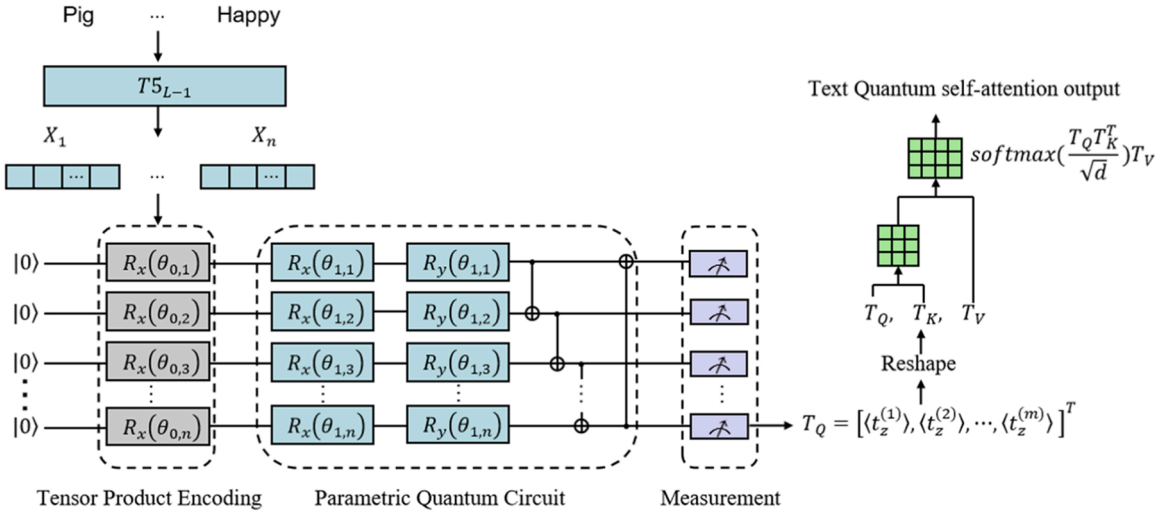


Fig. 3. Parameterizable Quantum Circuit Architecture Based on Text.

$$U_{enc}(H^{(L)}) = \prod_{i}^{\otimes} R_x(H^{(L)}, \theta) \tag{7}$$

$1 \le i \le n$, the encoded system state is:

$$\langle|y\rangle = U_{enc}(H^{(L)})\langle|0\rangle^{\otimes n} \tag{8}$$

where $H^{(L)}$ is the input data matrix, and $\langle|0\rangle^{\otimes n}$ represents n qubits all in the initial $\langle|0\rangle$ state. Next, the preliminarily encoded quantum states are further processed using parameterized quantum circuits, as shown in Fig. 3, primarily consisting of single-qubit rotation gates(R_x,R_y,etc.) and *CNOT* gates. Since the meaning of each word in a text often depends on its context, long-distance dependencies are particularly important in text processing. Therefore, rotation gates are used to enable each qubit to carry rich textual feature information. By training the rotation angles, the order and relative position of words can be accurately captured. Additionally, after qubits undergo *CNOT* operations, their states are no longer independent but form entangled states of multiple qubits. Through multiple *CNOT*operations, information can be transmitted across the entire quantum system. Although each operation involves only two qubits, multiple operations enable information sharing among all qubits, thus simulating long-distance dependencies in text and enhancing the model's semantic understanding.

$$U = \prod_{c}^{\otimes} R_x(c, \theta_{xc}) R_y(c, \theta_{yc}) CNOT[c, f] \tag{9}$$

$$f = \begin{cases} c+1 & c \neq N-1 \\ 0 & c = N-1 \end{cases} \tag{10}$$

represents the operation of the entire parameterized quantum circuit, c, $1 \le c \le n-1$, is the position of quantum bits, θ_{xc},θ_{yc} represent the angle parameters on $R_x$andR_y respectively. Then, initialize three quantum circuits U_q,U_k,U_v(U_q^+,U_k^+,U_v^+ correspond to their conjugate transposes, respectively) with corresponding angle parameters $\theta_q, \theta_k, \theta_v$ representing the query, key, and value parts, respectively. In the final stage, this paper measures the quantum states and apply the Pauli-Z operator (Z_i) to obtain the expected values of the qubits.

$$\langle t_q{}^i\rangle = \langle y|\rangle U_q^+(\theta_q) Z_i U_q(\theta_q)\langle|y\rangle \tag{11}$$

$$\langle t_k{}^i\rangle = \langle y|\rangle U_k^+(\theta_k) Z_i U_k(\theta_k)\langle|y\rangle \tag{12}$$

$$\langle t_v{}^i\rangle = \langle y|\rangle U_v^+(\theta_v) Z_i U_v(\theta_v)\langle|y\rangle \tag{13}$$

where $\langle t_q{}^i\rangle$,$\langle t_k{}^i\rangle$,$\langle t_v{}^i\rangle$ are the classical values obtained after measurement, respectively. Reshape the measured expected values into the form of our expected query (key and value) matrix. Thus, the $T_q = Reshape\left(\left[\langle t_q{}^1\rangle, \langle t_q{}^2\rangle, \cdots, \langle t_q{}^m\rangle\right]^T\right)(T_k, T_v)$ matrices are generated, equivalent to the Q, K, V matrices in the classical self-attention mechanism. Subsequently, the T_q, T_k, T_v matrices undergo the corresponding multi-head self-attention computation:

$$Attention(T_q, T_k, T_v) = softmax\left(\frac{T_q {T_k}^T}{\sqrt{d_k}}\right) T_v \tag{14}$$

$$MultiHead(T_q, T_k, T_v) = Concat(head_1, head_2, \cdots, head_h) W_O \tag{15}$$

where $\frac{T_q {T_k}^T}{\sqrt{d_k}}$ is the scaled dot product. $\sqrt{d_k}$acts as a scaling factor to ensure that the dot product values do not become too large, which could result in the softmax function's gradient vanishing. $head_i$ represents the attention computation for each head, and W_O is the weight matrix used to map the concatenated result back to the original feature space.

$$Z' = LayerNorm(H^{(0)} + MultiHead(T_q, T_k, T_v)) \tag{16}$$

$$FFN(Z') = Relu(Z' W_1 + b_1) W_2 + b_2 \tag{17}$$

$$Text_feaure = LayerNorm(Z' + FFN(Z')) \tag{18}$$

Finally, the result of the multi-head attention mechanism undergoes residual connection, layer normalization, and a feed-forward neural network (*FFN*) to obtain the text sequence feature *Text_feaure*.

3.3. Image feature extraction based on quantum vision transformer

This paper proposes an image-oriented parametric quantum circuit structure that, combined with the transformer, processes features extracted by the ResNet-50 model to extract deep image features. Unlike the quantum transformer used in text processing, different parametric quantum circuits are used in image feature extraction to adapt to the characteristics of image data.

The quantum circuit used in the quantum vision transformer is shown in Fig. 4. The structure of the quantum circuit used is similar to the quantum transformer in text feature extraction but is adjusted for specific implementations. The parameterized quantum circuit shown in Fig. 4 primarily consists of RBS (Reconfigurable Beam Splitter) gates. The RBS is a two-qubit parameterized gate, referred to as the reconfigurable beam splitter (RBS) in [30], defined as:

$$RBS(\theta) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos\theta & \sin\theta & 0 \\ 0 & -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \tag{19}$$

The RBS gate can be represented by a combination of basic gates, making it easier to simulate on classical computers.As shown in Fig. 5, H is the Hadamard gate,while $R_y(\theta)$ denotes the single-qubit rotation gate around the y-axis, and $\theta$is the rotation angle.The middle gate is the two-qubit CZ gate (Controlled-Z gate), also known as the controlled phase gate, primarily used to perform phase flips on the target qubit. Since the method in this paper simulates quantum circuits on classical computers, The depth and complexity of the quantum circuits need to be measured before designing them to ensure they can effectively extract image features.

The circuit structure shown in Fig. 4 primarily connects adjacent qubits through RBS gates, drawing inspiration from the pyramid circuit described in [31]. By training the parameters of the rotating gate in the RBS gate, it can better adapt to the data distribution of different images. And by applying RBS gates to adjacent two qubits, the final entangled state can contain global information in the image, which is very advantageous for image feature extraction. This paper chooses quantum circuits with relatively moderate circuit complexity and depth while ensuring model performance.

This paper conducted tests and comparisons using pyramid circuits and the quantum circuit proposed in this paper for image classification. Although pyramid circuits have more complex structures and deeper circuit depths, experimental results show that these two circuits perform equally well in our text classification task. This phenomenon can be analyzed from the following aspects:

Firstly, pyramid circuits attempt to introduce more complex entanglement relationships between qubits through multi-level RBS gate operations. However, in quantum image feature extraction tasks, the required entanglement level is not high. The parameterized quantum circuit used in this paper is sufficient to capture the main features in the data. This indicates that overly complex entanglement relationships have not significantly improved performance.

Secondly, the deeper the quantum circuit, the greater the effect of noise and error. Pyramid circuits can accumulate more noise and errors due to their complexity and deep circuit depth. Although pyramid circuits can theoretically introduce more entanglement, in practical

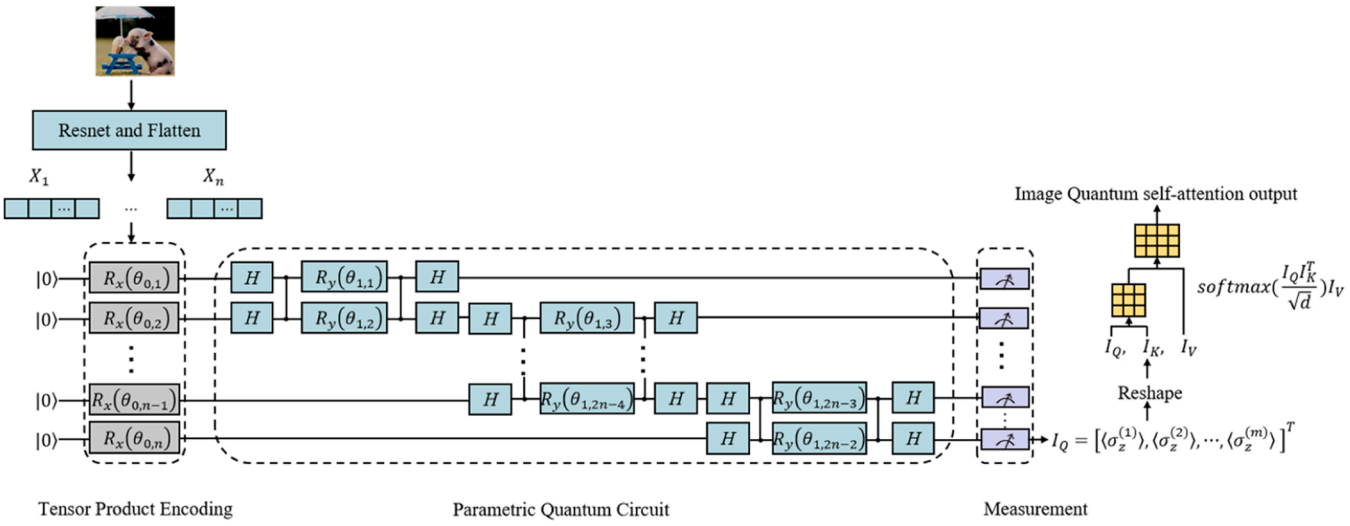


Fig. 4. Parameterizable Quantum Circuit Architecture Based on Image.

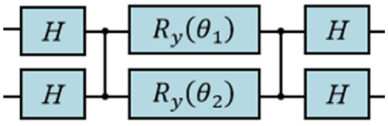


Fig. 5. A possible decomposition of the $RBS(\theta)$ gate.

operation the accumulation of noise and errors outweighs their potential advantages.

Finally, although pyramid circuits have complex structures, more quantum gate operations and deeper circuit depths require more computing resources. This paper proposes a detailed explanation in the performance analysis section of the model.

First, the input image $I \epsilon R^{h\times w\times c}$, where h is the height, w is the width, and c is the count of channels, is fed into the pre-trained ResNet-50 model. The ResNet-50 model is composed of numerous residual blocks, with each block containing multiple convolutional layers and batch normalization layers. The input image passes through the initial convolutional layer, batch normalization layer, and *ReLU* activation function to obtain the feature map F_1:

$$F_1 = ReLU(BN(Conv(I))) \tag{20}$$

where *BN* denotes batch normalization, and *ReLU* is the activation function. Then, the feature map F_1 is processed through multiple residual blocks. Each residual block has a skip connection between its input and output. For the i-th residual block, the output F_i can be expressed as:

$$F_i = F_{i-1} + ConvBlock_i(F_{i-1}) \tag{21}$$

$$ConvBlock(F) = BN(Conv(ReLU(BN(Conv(F))))) \tag{22}$$

where $ConvBlock_i$ represents the operations within the i-th residual block. After the last residual block, a global average pooling layer is applied to convert the final feature map F_{final} into a fixed-size vector:

$$F_{pool} = GlobalAveragePooling(F_{final}) \tag{23}$$

The spatial feature map $F_{pool} \in R^{h\times w\times c}$ after global average pooling is converted into sequence features $S \in R^{s\times h}$ through a flattening layer, where s represents the sequence length and h represents the number of hidden units.

$$S = Flatten(F_{pool}) = \{f_1, f_2, \cdots, f_s\} \tag{24}$$

where f_i represents the i-th flattened feature vector. The flattened sequence features S are then fed into the quantum image transformer for processing, starting with positional encoding:

$$t = PositionalEncoding(S) \tag{25}$$

The position-encoded image features $t = \{i_0, i_1, \cdots, i_n\}, i_l \in R^d$, $0 \le l \le n$(where d represents the embedding dimension) are input into the quantum transformer. The process is similar to the quantum transformer used for text feature extraction. This paper uses the quantum multi-head attention mechanism, replacing the weight matrices in the attention mechanism with quantum circuits. First, the position-encoded image features are converted into quantum states, ensuring that the quantum circuits can process this information.

$$\langle|v\rangle = U_{enc}(t)\langle|0\rangle^{\otimes n} \tag{26}$$

where $\langle|v\rangle$ represents the state after encoding. Then comes the parameterized quantum circuit part for images.

$$U = \prod_j^{\otimes} H(j)H(j+1)CZ[j,j+1]R_y(j,\theta_{2j-1})R_y(j+1,\theta_{2j})CZ[j,j+1]H(j)H(j+1) \tag{27}$$

$1 \le j \le n-1$, U represents the operation of the entire parameterized quantum circuit. Then, initialize three quantum circuits U_q,U_k,U_v (U_q^+,U_k^+,U_v^+correspond to their conjugate transposes, respectively) and their corresponding angle parameters θ_q,θ_k,θ_v. Finally, measurements are taken, where the Pauli-Z operator (Z_i) is used to obtain the expected value of qubits.

$$\langle\sigma_q{}^i\rangle = \langle v|U_q^+(\theta_q)Z_iU_q(\theta_q)\langle|v\rangle \tag{28}$$

$$\langle\sigma_k{}^i\rangle = \langle v|U_k^+(\theta_k)Z_iU_k(\theta_k)\langle|v\rangle \tag{29}$$

$$\langle\sigma_v{}^i\rangle = \langle v|U_v^+(\theta_v)Z_iU_v(\theta_v)\langle|v\rangle \tag{30}$$

where $\langle\sigma_q{}^i\rangle$,$\langle\sigma_k{}^i\rangle$,$\langle\sigma_v{}^i\rangle$ are the classical values obtained after measurement, respectively. After multiple measurements, classic data is obtained and then reshaped into the desired shape to form a query matrix $I_q = Reshape\left([\langle\sigma_q{}^1\rangle, \langle\sigma_q{}^2\rangle, \cdots, \langle\sigma_q{}^m\rangle]^T\right)$ and obtain the key matrix I_k and value matrix I_v through the same steps. The next step is to perform the corresponding multi-head self-attention computation:

$$Attention(I_q, I_k, I_v) = softmax\left(\frac{I_q {I_k}^T}{\sqrt{d_k}}\right) I_v \tag{31}$$

$$MultiHead(I_q, I_k, I_v) = Concat(head_1, head_2, \cdots, head_h) W_O \tag{32}$$

where $\frac{I_q {I_k}^T}{\sqrt{d_k}}$ is the scaled dot product. $\sqrt{d_k}$acts as a scaling factor to ensure that the dot product values do not become too large, which could result in the softmax function's gradient vanishing. $head_i$ represents the attention computation for each head, and W_O is the weight matrix used to map the concatenated result back to the original feature space.

$$Z' = LayerNorm(t + MultiHead(I_q, I_k, I_v)) \tag{33}$$

$$FFN(x) = Relu(Z' W_1 + b_1) W_2 + b_2 \tag{34}$$

$$Image_feaure = LayerNorm(Z' + FFN(Z')) \tag{35}$$

Then, the features are routed through a feed-forward neural network, along with residual connections and normalization, producing the *Image_feaure*.

3.4. Text-image alignment

To address the issue of cross-modal semantic association difficulties caused by differing feature distributions between text and visual modalities, this paper introduces a contrastive learning method [32]. The goal is to enhance the quality of unimodal representations prior to performing feature fusion, thereby making the modal representations more consistent. In a batch, we include M image-text pairs (V,T), where V represents image data. For each text T, the corresponding image V is treated as a positive sample, while the remaining M−1 images in the batch are considered negative samples. The detailed process is as follows:

Firstly, we use multi-layer perceptron (MLP) layers $j_v(\bullet)$ and $j_t(\bullet)$ to map the [CLS] embedding vectors $v_{cls} \epsilon R^{1\times d}$ and $t_{cls} \epsilon R^{1\times d}$ to normalized low-dimensional vector representations [33]. The [CLS] embedding vectors are obtained from the *Text_feaure* and *Image_feaure*. By transforming the [CLS] embedding vectors through MLP layers, better results can be achieved compared to simple linear transformations. Then, we define the cosine similarities $s(V,T) = j_v(v_{cls})^T j_t(t_{cls})$ and $s(T,V) = j_t(t_{cls})^T j_v(v_{cls})$ between text and images to compute the similarity between images and text within a batch. Next, we compute the similarity from image to text and text to image within the batch as follows:

$$q^{i2t}(V) = \frac{\exp(s(V,T)/\tau)}{\sum_{i=1}^{M} \exp(s(V,T_i)/\tau)}, q^{t2i}(T) = \frac{\exp(s(T,V)/\tau)}{\sum_{i=1}^{M} \exp(s(T,V_i)/\tau)} \tag{36}$$

where τ is a learnable temperature parameter. Let $y^{i2t}(V)$ and $y^{t2i}(T)$ represent the true one-hot similarities.The contrastive loss is defined as the cross-entropy H between q and y:

$$\ell = H\left(y^{i2t}(V), q^{i2t}(V)\right) + H\left(y^{t2i}(T), q^{t2i}(T)\right) \tag{37}$$

3.5. Feature fusion

Generally, the obtained text and image features contain a lot of content unrelated to emotional information. If the interaction between modalities is overlooked, the resulting multimodal features are bound to contain a considerable amount of redundant information that is irrelevant to the emotional content. Therefore, this paper proposes a cross-attention fusion framework, in which the attention mechanism is implemented on the representations of both modalities. First, the text features are used as queries and compared with the image features to obtain a feature representation that integrates both text and image information. This process helps the model extract the most relevant emotional information from the images based on the text. Then, the image features are used as queries and compared with the text features to obtain a feature representation that integrates both image and text information. This process helps the model extract the most relevant emotional information from the text based on the images.

The cross-attention fusion calculation processes are as follows:

From Section 3, we obtain text feature $T: = Text_feaure \in R^{l_t \times d_t}$ and$I: = Image_feaure \in R^{l_i \times d_i}$.

The text and image features are mapped to the same dimension d through linear transformations:

$$T^{`} = TW_T + b_T and I^{`} = IW_I + b_I \tag{38}$$

where $W_T \in R^{d_t \times d}$ and $W_I \in R^{d_i \times d}$ are the linear transformation matrices, and b_T and b_I are the bias terms. Next, generate queries, keys, and values for each modality:

$$Q_T = T^{`} W_Q^T, K_T = T^{`} W_K^T, V_T = T^{`} W_V^T \tag{39}$$

$$Q_I = I^{`} W_Q^I, K_I = I^{`} W_K^I, V_I = I^{`} W_V^I \tag{40}$$

where $W_Q^T, W_K^T, W_V^T \in R^{d \times d_k}$ and $W_Q^I, W_K^I, W_V^I \in R^{d \times d_k}$are the linear transformation matrices for queries, keys, and values, and d_k is the dimension in the attention mechanism.

Calculate the attention score between each text and image feature:

$$A_{T \to I} = softmax(\frac{Q_T K_I^\top}{\sqrt{d_k}}) \tag{41}$$

$$A_{I \to T} = softmax(\frac{Q_I K_T^\top}{\sqrt{d_k}}) \tag{42}$$

where $A_{T \to I} \in R^{L_t \times L_i}$ and $A_{I \to T} \in R^{L_i \times L_t}$ represent the attention scores of text features on image features and image features on text features, respectively. Then, use the attention scores to weight the values, obtaining the fused features:

$$T_{fused} = A_{T \to I} V_I \tag{43}$$

$$I_{fused} = A_{I \to T} V_T \tag{44}$$

where $T_{fused} \in R^{L_t \times d}, I_{fused} \in R^{L_i \times d}$ represent the weighted text features and weighted image features obtained after cross-attention fusion, respectively.

Finally, this paper achieve further alignment and fusion of T_{fused} and I_{fused} through the Transformer Encoder. This approach can further capture the fine-grained relationships between text and image features, improving the performance of multimodal sentiment analysis.

$$F = Concat([T_{fused}, I_{fused}]) \tag{45}$$

where $F \in R^{(L_t + L_i) \times d}$ represents the concatenated result of T_{fused}andI_{fused}.The concatenated result F is then fed into the Transformer Encoder model for processing. It can capture the global dependencies of the sequence through the self-attention mechanism,thus achieving feature alignment and fusion:

$$F_{aligned} = TE_{\times N}(F) \tag{46}$$

where $F_{aligned} \in R^{(L_t + L_i) \times d}$ represents the aligned features processed through N layers of the Transformer Encoder.

To generate the final emotion prediction, the fused feature vector $F_{aligned}$ is fed into the softmax layer:

$$y = softmax(W_f F_{aligned} + b_f) \tag{47}$$

where W_f and b_f are learnable parameters. Our model is optimized using the AdamW optimizer to minimize the cross-entropy loss function:

$$\mathscr{L} = -\sum_i \hat{y}_i \log y_i \tag{48}$$

where $\hat{y}_i$ represents the true emotional label, while y_i is the output of the softmax layer.

The final loss function $\varkappa$ based above loss function is as follows:

$$\varkappa = \alpha\mathscr{L} + \beta\mathscr{L} \tag{49}$$

where α,β are hyperparameters to balance their contributions to the overall loss.

4. Experiments

4.1. Dataset

This paper validated our proposed method on the MVSA and HFM datasets. The MVSA dataset is commonly used for multimodal sentiment analysis tasks. It consists of two parts: MVSA-Single and MVSA-Multiple. The MVSA-Single dataset contains 5129 image and text pairs, while the MVSA-Multiple dataset is larger, containing 19,600 image and text pairs. Both the MVSA and HFM datasets are collected from Twitter, where each data entry includes an image and text pair. The MVSA and HFM datasets were split randomly into training, validation, and test sets using an 8:1:1 ratio, with the statistical details shown in Table 1.

4.2. Implementation details

This section discusses the specific details of the experimental setup for the proposed method. All experiments were implemented on an NVIDIA P100 GPU using Python 3.10 and the Flax framework, with the TensorCircuit library used to construct quantum circuits and simulate the quantum environment. For the MVSA-Single, MVSA-Multiple, and HFM datasets, since the quantum environment was simulated on a classical computer, which is computationally intensive, the batch sizes were set to 8, 2, and 2, respectively. To ensure a fair comparison, the same preprocessing method as in [49] was applied to the two MVSA datasets, while the preprocessing for HFM followed the approach in [50]. In addition, these three datasets were randomly divided into training, validation, and test sets in an 8:1:1 ratio. We used the AdamW optimizer with a learning rate of 2e-5. The specific parameter settings are shown in Table 2. The models were evaluated using accuracy and F1-score, defined as follows:

$$Precision = \frac{TP}{TP + FP} \tag{47}$$

$$Recall = \frac{TP}{TP + FN} \tag{48}$$

$$F1 = \frac{2 \bullet Precision \bullet Recall}{Precision + Recall} \tag{49}$$

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{50}$$

where *TP* is true positive, *TN* is true negative, *FP* is false positive and *FN* is false negative.

In this experiment, a cosine annealing learning rate schedule with warm-up is used, implemented through the Optax library. **Lrs_peak** represents the maximum learning rate reached after the warm-up phase. **Lrs_warmup** indicates the number of warm-up steps, controlling the total number of steps for the learning rate to increase linearly from 0 to the peak value. **Lrs_decay** refers to the number of steps over which the learning rate decays from the peak value to 0, following a cosine function.

Table 1
Statistics of the MVSA and HFM datasets.

Dataset	Train	Val	Test	Total
MVSA-Single	3611	450	450	4511
MVSA-Multiple	13,642	2409	2410	17,024
HFM	19,816	2410	2409	24,635

4.3. Compared methods

This paper conducted a contrast between our method and baseline methods in both unimodal and multimodal settings.

4.3.1. Unimodal baselines model

For the text modality, widely recognized models for text classification tasks include CNN [34] and Bi-LSTM [35]. TGNN [36] is a graph neural network at the text level, designed for text classification tasks. BERT, developed by Devlin et al. (2019)[37], is a pre-trained language model optimized for sentiment analysis, using bidirectional context to enhance prediction accuracy. OSDA [38] is a multi-view model designed for image sentiment analysis, whereas ResNet [39] is a pre-trained image model that can be adapted and fine-tuned for a range of tasks.

4.3.2. Multimodal baselines model

MultiSentiNet [40] uses an attention mechanism to perform multimodal sentiment analysis. A hierarchical semantic attention network based on image descriptions is used by HSAN [41] to perform multimodal sentiment analysis. Co-MN-Hop6 [42] integrates a joint memory network to iteratively capture interactions between different modalities. MGNNS [43] is an emotion-aware multi-channel graph neural network, specifically designed to detect sentiment in image-text data. Feature vectors from multiple modalities were fused to create a multimodal representation [44]. Concat(2) involves the concatenation of text and image features, whereas Concat(3) involves the addition of an image attribute feature. MMSD [45] integrates text, images and image attributes using a multimodal hierarchical fusion model. [46] proposed D&R Net, which combines these modalities by constructing a decomposition and relationship network. CLMLF[47] developed a multi-level fusion model that uses BERT and ResNet, and employs a multi-layer Transformer-Encoder to align and fuse multimodal features, thereby improving performance in multimodal tasks.[48] introduces Cross-modal Translation Based Meta Weight Adaption (CTMWA) for Robust Image-Text sentiment analysis. The approach employs a cross-modal translation network to capture shared concepts between text and image content, enabling effective handling of missing modalities. A unimodal weight adaption strategy, based on meta-learning, adjusts the modality encoder weights to improve performance. CiteNet [49] proposes a cross-modal incongruity perception network designed for multimodal sentiment prediction, aiming to address sentiment incongruence between different modalities. To achieve this, the researchers introduce the CiteNet model, which includes three main modules: cross-modal alignment, cross-modal fusion, and cross-modal incongruity learning. These modules produce incongruity scores, which are then used to dynamically fuse unimodal and multimodal representations, leading to improved sentiment prediction accuracy. [50] proposed a Context-Aware Dual Attention Network for Multimodal Sarcasm Detection that simultaneously processes text and image attention mechanisms through a shared memory module to identify sarcastic sentiments. It also leverages the implicit context of a multimodal commonsense graph to integrate semantic and emotional information and employs cross-modal matching techniques to detect contradictions between text and images, ultimately enhancing the accuracy of sarcasm recognition. [51] proposed a Dual-Branch Dynamic Routing Network that utilizes a dynamic routing network to capture affective cues from different modalities for multimodal sentiment detection.

In this study, the experimental results of the baseline models are directly from the relevant references. The method proposed in this paper

Table 2
Table with various datasets parameters.

Dataset	Epoch	Lrs_peak	Lrs_warmup	Lrs_decay	Batch_size	Dropout	GPU
MVSA-Single	30	0.001	500	4500	8	0.2	P100
MVSA-Multiple	30	0.001	40,000	360,000	2	0.2	P100
HFM	60	0.001	60,000	540,000	2	0.2	P100

was quantitatively compared with the baseline models in three datasets, as shown in Table 3. In all three datasets, most of the multimodal sentiment analysis models outperformed the unimodal models; clearly, the introduction of multimodal data provides more effective information for multimodal sentiment analysis tasks. First, in the multimodal sentiment analysis task, the proposed method in this paper achieved higher accuracy and F1 scores than other methods on the MVSA-Single dataset. For example, compared to the CTMWA method, it improved accuracy by 1.22 % and F1 score by 0.67 %, and compared to the CiteNet method, it improved accuracy by 1.04 % and F1 score by 1.74 %. On the MVSA-Multiple dataset,the proposed method in this paper outperformed the CTMWA method by 1.54 % in accuracy and 1.45 % in F1 score, and outperformed CiteNet by 2.67 % in accuracy and 4.94 % in F1 score. In addition, because the HFM dataset is a binary classification task with sentiments simplified into positive and negative categories, some complex emotional expressions may not be accurately recognized, which could affect the improvement of metrics and result in smaller improvements in model performance. Nevertheless, the method proposed in this paper achieves accuracy similar to that of the D^2R method. and increased the F1 score by 1.38 % compared to the CADAN method.

The superior performance of the proposed method is mainly attributed to the following three aspects:

First, the Hilbert space constructed based on quantum theory provides a multidimensional representation that allows emotional information (including emotional expressions in the form of words, phrases, or images) to be mapped into a complex vector space. In this space, the quantum superposition characteristics enables emotional information to exist simultaneously in multiple possible states, effectively expressing the uncertainty and ambiguity within emotional data. This is especially important in scenarios where emotions are intertwined and difficult to distinguish, as quantum superposition can capture subtle emotional differences that traditional models struggle to identify. Through comparative experiments, we validated the critical impact of quantum superposition characteristics on emotional analysis tasks.

Table 3
Experimental results of models on MVSA and HFM.

Modality	Model	MVSA-Single	MVSA-Multiple	Model	HFM
		ACC F1	ACC F1		ACC F1
Text	CNN[34]	0.6819 0.5590	0.6564 0.5766	CNN[34]	0.8003 0.7532
	BiLSTM[35]	0.7012 0.6506	0.6790 0.6790	BiLSTM[35]	0.8190 0.7753
	BERT[37]	0.7111 0.6970	0.6759 0.6624	BERT[37]	0.8389 0.8326
	TGNN[36]	0.7034 0.6594	0.6967 0.6180	N/A	N/A N/A
Image	ResNet−50 [39]	0.6467 0.6155	0.6188 0.6098	ResNet−50 [39]	0.7277 0.7138
	OSDA[38]	0.6675 0.6651	0.6662 0.6623	ResNet−101 [39]	0.7248 0.7122
Multimodal	MultiSentiNet [40]	0.6984 0.6894	0.6886 0.6711	Concat(2) [44]	0.8103 0.7799
	HSAN[41]	0.6988 0.6690	0.6796 0.6776	Concat(3) [44]	0.8174 0.7874
	Co-MN-Hop [42]	0.7051 0.7001	0.6892 0.6883	MMSD[43]	0.8344 0.8018
	MGNNS[43]	0.7377 0.7270	0.7249 0.6934	D&R Net [46]	0.8402 0.8060
	CLMLF[47]	0.7533 0.7346	0.7200 0.6983	CLMLF[47]	0.8543 0.8487
	CTMWA[48]	0.7591 0.7574	0.7402 0.7384	CADAN[50]	0.8693 0.8374
	CiteNet[49]	0.7609 0.7467	0.7289 0.7035	N/A	N/A N/A
	D^2R[51]	0.7667 0.7559	0.7159 0.7058	D^2R[51]	0.8672 **0.8625**
	Ours	**0.7713** **0.7641**	**0.7556** **0.7529**	**Ours**	**0.8697** 0.8512

The ITQT-MSA-NoH model in Table 4 refers to the ITQT-MSA model with the H gate removed from the proposed parameterized quantum circuit. Since the quantum superposition characteristic is achieved through the H gate, removing the H gate prevents the model from generating superposition states, resulting in the inability of emotional information to exist in multiple states simultaneously. Experimental results show that the ITQT-MSA-NoH model exhibits a significant decrease in both accuracy and F1 score in emotional analysis tasks, specifically with accuracy dropping from 0.7713 to 0.7498 and the F1 score decreasing from 0.7641 to 0.7256. This result further demonstrates the important role of quantum superposition characteristics in enhancing multimodal emotional analysis tasks. Additionally, due to the curse of dimensionality, classical auto-encoders perform poorly when handling high-dimensional data, limiting their ability to efficiently compress and reconstruct data. Quantum superposition, on the other hand, can encode information simultaneously across multiple states, enabling a more efficient representation of high-dimensional data. This feature allows quantum models to handle complex and high-dimensional emotional data better than traditional methods. It provides more flexible and precise emotional analysis capabilities.

Second, quantum entanglement plays a crucial role in quantum-inspired models. The entanglement characteristic allows for the establishment of global associations between emotional features, enhancing the model's understanding of emotional information. Specifically, through quantum entanglement, different emotional information is no longer processed independently but is interconnected through parameters in the quantum circuit, thereby combining local emotional data with the global emotional context. This ability to capture global associations helps the model better recognize the complex relationships between emotional features in emotional analysis, significantly improving the model's accuracy and expressiveness. Through comparative experiments, we validated the critical impact of quantum entanglement characteristics on emotional analysis tasks.

The ITQT-MSA-NoCNOT model in Table 4 refers to the ITQT-MSA model with the CNOT gate removed from the proposed parameterized quantum circuit. Since the quantum entanglement characteristic is achieved through the CNOT gate, removing the CNOT gate prevents the model from effectively capturing the global associations of emotional features, thereby affecting the overall emotional analysis capability. Experimental results show that the ITQT-MSA-NoCNOT model's accuracy drops to 0.7201 and the F1 score decreases to 0.7125, representing reductions of approximately 5.12 % and 5.16 % respectively compared

Table 4
Experimental results of ITQT-MSA without H and CNOT gates.

Model	ACC	F1
ITQT-MSA-NoCNOT	0.7201	0.7125
ITQT-MSA-NoH	0.7489	0.7256
ITQT-MSA	**0.7713**	**0.7641**

to the ITQT-MSA model. This significant performance decline further validates the crucial role of quantum entanglement characteristics in multimodal emotional analysis tasks.

Finally, during the feature fusion stage, this paper employs a cross-attention fusion mechanism to integrate text and image features. This mechanism introduces attention weights between different modalities, allowing the selection and focus on the most critical features for sentiment analysis. In particular, words in the text modality can be weighted based on visual features from the image modality, and vice versa. This cross-attention fusion effectively captures the correlations between features from different modalities, making feature fusion more precise. In addition, the cross-attention fusion mechanism allows for detailed interactions between text and image features. For example, specific image regions may correlate highly with certain text words, and the cross-attention fusion mechanism captures this fine-grained correlation, enhancing sentiment analysis accuracy.

In conclusion, our proposed quantum transformer-based multimodal sentiment analysis framework performs excellently across multiple datasets, validating its effectiveness and superiority in multimodal sentiment analysis tasks.

5. Model performance analysis

5.1. Ablation experiments

5.1.1. Performance comparison between quantum and classical methods

In this paper, an additional evaluation of the effects of the cross-attention fusion module was conducted alongside the quantum transformer-based method for classifying image-text sentiment. The results of these evaluations are shown in Table 5.

This paper performed sentiment analysis experiments on single modalities. For the text modality, the last transformer layer of the pre-trained T5 model was replaced with a quantum transformer for text, and a comparison was made with the original T5 model. The results from the three datasets show that the quantum transformer significantly improved model performance, indicating its superior capability in capturing complex relationships and features in the text.

For the image modality, this study replaced the transformer module in the hybrid VIT model with an image-based quantum transformer. The experimental results demonstrate the incorporation of the image-based quantum transformer improved performance on all three datasets, with particularly significant gains on the MVSA-Single dataset. This further confirms the potential advantages of the quantum transformer in capturing complex image features and enhancing image sentiment classification.

Within the entire multimodal sentiment analysis framework, the incorporation of quantum transformers also led to a significant enhancement in overall model performance. In the ITCT-MSA model, image feature extraction was performed using a combination of ResNet-50 and a classical transformer, while text feature extraction used the pre-trained T5 model, and cross-attention feature fusion was implemented. In the ITQT-MSA model, the classical transformer used for image feature extraction was replaced by an image-based quantum transformer, and the final transformer layer of the T5 model was replaced by a quantum transformer for text. With these modifications, ITQT-MSA outperformed ITCT-MSA in terms of accuracy (ACC) on the MVSA datasets and achieved a lower loss value. Specifically, on the MVSA-Single dataset, as shown in Fig. 6, ITQT-MSA achieved an accuracy of 0.7713 and a loss of 0.0025, representing an improvement of 1.12 % in accuracy and a reduction of 0.05 % in loss compared to ITCT-MSA. Furthermore, on the MVSA-Multiple datasets, ITQT-MSA also demonstrated consistent performance improvements, with a final accuracy of approximately 0.75, and the loss remained at a low level, indicating that the introduction of the quantum transformer enhanced the model's generalization ability and training stability. These experimental results clearly demonstrate that quantum transformers can effectively enhance feature extraction and fusion capabilities in both single-modal and multimodal sentiment classification tasks, leading to significant improvements in overall model performance.

Table 5
Ablation results of our ITQT-MSA.

Modality	Model	MVSA-Single		MVSA-Multiple	
		ACC	F1	ACC	F1
Text	T5	0.7198	0.7070	0.6851	0.6712
	$T5_{L-1}$+QTR	**0.7222**	**0.7201**	**0.7001**	**0.6979**
Image	ResNet−50 +TR	0.6467	0.6155	0.6188	0.6098
	ResNet−50 +QTR	**0.6675**	**0.6651**	**0.6389**	**0.6290**
Multimodal	ITCT-MSA	0.7601	0.7511	0.7363	0.7226
	ITQT-MSA	**0.7713**	**0.7641**	**0.7556**	**0.7529**

5.1.2. Performance comparison of quantum circuit structures

This paper compares two different parametric quantum circuit structures: the pyramid circuit and the circuit this paper used. Table 6 presents the performance comparison of these circuits in terms of circuit depth, the number of quantum gates, classification accuracy, F1 score, and time complexity. From this comparison, this paper concludes that for practical multimodal sentiment analysis tasks, the circuit proposed in this paper can achieve classification performance similar to that of complex circuits while maintaining lower computational complexity and resource consumption, demonstrating higher practical value.

5.1.3. Analysis of resource consumption and practical value of quantum methods

In order to demonstrate the practicality of the proposed method in resource-constrained environments, we conducted a detailed analysis of the differences between the ITQT-MSA method and baseline methods in terms of model training time and GPU memory usage. As shown in Table 7, quantum-inspired models consume significantly more time and resources compared to classical models. The primary reason for this increased consumption is that quantum systems utilize superposition and entanglement, causing the state space to grow exponentially with the number of qubits. This exponential growth demands substantial computational resources.

Although quantum-inspired models are more resource-intensive, exploring them in the context of sentiment analysis can significantly advance interdisciplinary research between artificial intelligence and quantum computing. This cross-disciplinary exploration not only has the potential to lead to new theoretical breakthroughs but also may foster the development of innovative algorithms and technologies, thereby driving the progress of the entire field.

5.2. Visualization

5.2.1. Visualization of image features

This paper visualized and compared the text and image features



Fig. 6. The Acc and Loss of ITCT-MSA and ITQT-MSA on the MVSA-Single and MVSA-Multiple datasets, respectively.

Table 6
Comparison of different quantum circuit architectures.

Circuit	Depth	Gates	ACC	F1	Time complexity
Pyramid	$2N-3$	$N(N-1)/2$	0.7716	0.7650	$O(N^2)$
Our	$N-1$	$N-1$	0.7713	0.7641	$O(N)$

obtained by the quantum transformer, as shown in Fig. 7. In Fig. 7(a), the image features extracted by ResNet-50 +Transformer show attention focused on the central area of the image, especially on the face and head of the central figure. In Fig. 7(b), the features extracted by ResNet-50 +Quantum Transformer display a more uniform attention distribution. In addition to the central figure, the faces of other figures also receive significant attention. This indicates that the quantum transformer has a better ability to perceive global information.

5.2.2. *Visualization of image features*

The following visualizations show the attention weights of the first head of the transformer in the last layer of the T5 model for text feature extraction. As shown in Fig. 8, Fig. 8(a) represents the attention weights obtained using classical attention, while Fig. 8(b) shows the attention weights obtained using quantum attention. From these two figures, it can be seen that the first image focuses more on the beginning of the sentence, especially the subject. In contrast, the second image shows attention across different parts of the sentence. This suggests that the quantum attention mechanism is more effective at capturing the overall information of the sentence.

5.2.3. *Visualization comparison of cross-attention and self-attention*

To validate the advantage of our framework in modeling the relevant regions between text and images, we visualized the attention weights of two samples in Fig. 9. When specific keywords are provided, cross-modal attention can identify targets in the images more effectively and assign them higher attention weights compared to self-attention. As shown in Fig. 9(a), when the word "sandwiches" is marked, cross-modal attention focuses on the image regions displaying "sandwiches," whereas self-attention simultaneously focuses on regions displaying both "books" and "sandwiches." In Fig. 9(b), when the word "dogs" is marked, cross-modal attention concentrates solely on the image regions displaying "dogs," while self-attention simultaneously focuses on regions displaying both "people" and "dogs." This demonstrates that our method can accurately align textual words with image regions and reduce interference from irrelevant objects, thereby playing a crucial role in the fusion stage of text and image features.

5.3. *Case analysis*

To provide a more comprehensive assessment of how the ITQT-MSA model performs, this paper randomly selected four samples from the dataset and conducted experiments with both ITQT-MSA and CiteNet models. Table 8 shows an image example of the selected sample, and Table 8 displays the prediction results of two models for each sample

Table 7
The training time and GPU memory usage of ITQT-MSA with baselines.

Modality	Model	MVSA-Single		MVSA-Multiple	
		Training Time(hours)	GPU Memory(G)	Training Time(hours)	GPU Memory(G)
Text	T5	1	9.6	1.2	10.5
	$T5_{L-1}$+QTR	1.5	9.8	2	11.2
Image	ResNet−50 +TR	1.5	12.5	2	11
	ResNet−50 +QTR	2.5	12.6	3	12.3
Multimodal	ITCT-MSA	3	13.3	4	14
	ITQT-MSA	7	13.5	8	15.6



Fig. 7. Visualization results of classical attention weights and quantum attention weights in images selected from tweets.

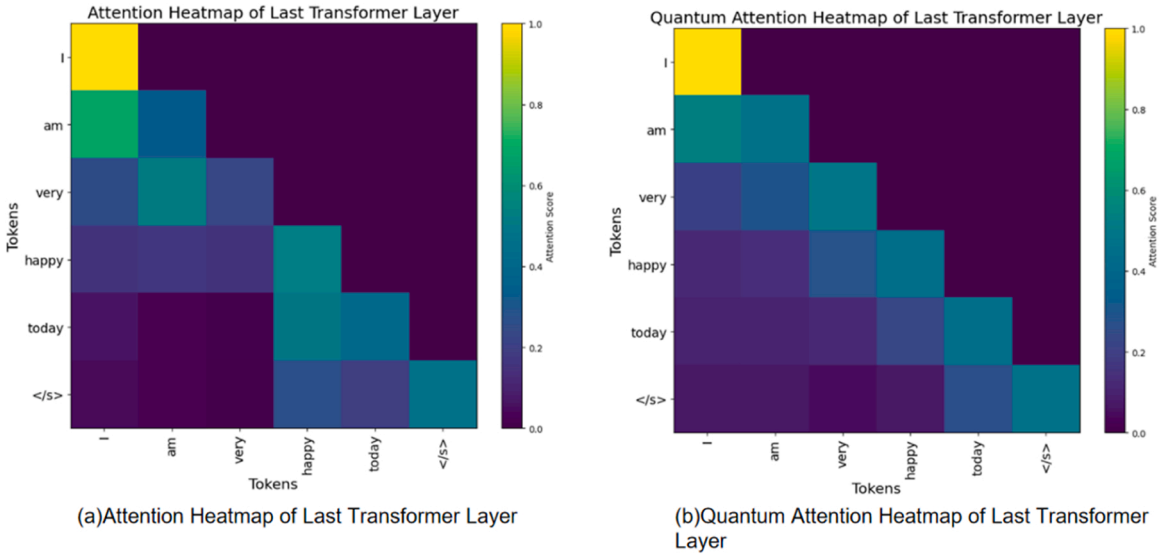


Fig. 8. Visualization results of classical attention weights and quantum attention weights in text selected from tweets.

(text image pair), with correct predictions highlighted in black font and incorrect predictions highlighted in red font. As shown in Table 8, the CiteNet model made an incorrect prediction for the last sample in the table. This error could be due to the model excessively magnifying irrelevant image details, leading to inaccurate sentiment interpretation of the text. The classification results show that CiteNet correctly predicted three out of the four selected samples, while ITQT-MSA correctly predicted all four, achieving a 100 % accuracy rate. The results confirm the effectiveness of the ITQT MSA model in carrying out a multi-modal sentiment analysis.

6. Conclusion

This paper proposes a quantum transformer-based framework(ITQT-MSA), specifically designed for image and text. This framework includes quantum text and image feature extraction modules, and a cross-attention fusion module. In the image and text feature extraction modules, this paper uses the pre-trained models T5 and ResNet-50 combined with quantum transformers and employ different parameterized quantum circuits to process information from different modalities, thereby extracting deeper features from both text and images. Then, the cross-attention fusion module integrates the extracted text and image features, and a classifier makes the final sentiment category decision. Experimental results on the large-scale datasets MVSA and HFM demonstrate that our method significantly outperforms some of the latest sentiment analysis algorithms. This result not only demonstrates the potential of quantum computing in traditional Natural Language Processing (NLP) and Computer Vision (CV) fields, but also provides new ideas and directions for addressing some of the challenges faced by existing algorithms. Quantum computing, with its unique computational

capabilities, can outperform classical computing methods when dealing with high-dimensional, complex data, thus providing strong support for the further development of fields such as sentiment analysis, automated text generation, and multimodal data analysis.

In terms of potential applications, our approach holds significant theoretical value and can be widely applied in various practical scenarios. For example, it can greatly enhance the accuracy of sentiment understanding and system responsiveness in fields such as social media data analysis, user behavior prediction, intelligent customer service, and emotional interaction systems. Furthermore, with the continuous advancement of quantum computing technology, this framework could be applied to more complex multimodal data analysis tasks in the future, driving innovation in intelligent decision-making systems.

Future work will include other modalities such as audio in sentiment analysis. Additionally, we will explore optimizing the computational efficiency and resource utilization of the quantum computing components within the constraints of current computing resources and will explore hybrid models combining quantum and classical computing to leverage their respective advantages.

Algorithm 1. : Quantum Text Feature Extraction

Data: Text data $X = \{x_1, x_2, \cdots, x_n\}$
Result: Text_feature
Initialization:
For $x \leftarrow x_1$ **to** x_n **do**:
    $T \leftarrow \text{SentencePiece}(x)$
    $E \leftarrow \text{Embedding}(T)$
    $H^{(0)} \leftarrow E + \text{PositionalEncoding}(T)$
    For $i \in [1, L]$ **do**
        $H^{(i)} \leftarrow \text{Transformer}(H^{(i-1)})$
    End for
    Initialize Quantum State:
    $|\psi\rangle \leftarrow U_{enc}(H^{(L)})|0\rangle^{\otimes n}$
    Quantum Operation And Measurement:
    $\langle t_q^i \rangle \leftarrow \langle \psi | U_q^T(\theta_q) Z_i U_q(\theta_q) | \psi \rangle$
    $\langle t_k^i \rangle \leftarrow \langle \psi | U_k^T(\theta_k) Z_i U_k(\theta_k) | \psi \rangle$
    $\langle t_v^i \rangle \leftarrow \langle \psi | U_v^T(\theta_v) Z_i U_v(\theta_v) | \psi \rangle$
    $T_q \leftarrow \text{Reshape}([\langle t_q^1 \rangle, \langle t_q^2 \rangle, \cdots, \langle t_q^m \rangle]^T)$
    $T_k \leftarrow \text{Reshape}([\langle t_k^1 \rangle, \langle t_k^2 \rangle, \cdots, \langle t_k^m \rangle]^T)$
    $T_v \leftarrow \text{Reshape}([\langle t_v^1 \rangle, \langle t_v^2 \rangle, \cdots, \langle t_v^m \rangle]^T)$
    Extract Text Features:
    $\text{Attention}(T_q, T_k, T_v) \leftarrow \text{softmax}(\frac{T_q {T_k}^T}{\sqrt{d_k}}) T_v$
    $\text{MultiHead}(T_q, T_k, T_v) \leftarrow \text{Concat}(head_1, head_2, \cdots, head_h) W_0$
    $Z' \leftarrow \text{LayerNorm}(H^{(0)} + \text{MultiHead}(T_q, T_k, T_v))$
    $\text{FFN}(Z') \leftarrow \text{ReLU}(Z' W_1 + b_1) W_2 + b_2$
    $\text{Text_feature} \leftarrow \text{LayerNorm}(Z' + \text{FFN}(Z'))$
End for
Return Text_feature

Algorithm 2. : Quantum Image Feature Extraction

Data: Image data $I = \{i_1, i_2, \cdots, i_n\}$
Result: Image_feature
Initialization:
For $i \leftarrow i_1$ **to** i_n **do**:
    $T \leftarrow$ ResNet50(i)
    $t \leftarrow T$ + PositionalEncoding(T)
    Initialize Quantum State:
    $|\psi\rangle \leftarrow U_{enc}(t)|0\rangle^{\otimes n}$
    Quantum Operation And Measurement:
    $\langle\sigma_q^i\rangle \leftarrow \langle\psi|U_q^T(\theta_q)Z_iU_q(\theta_q)|\psi\rangle$
    $\langle\sigma_k^i\rangle \leftarrow \langle\psi|U_k^T(\theta_k)Z_iU_k(\theta_k)|\psi\rangle$
    $\langle\sigma_v^i\rangle \leftarrow \langle\psi|U_v^T(\theta_v)Z_iU_v(\theta_v)|\psi\rangle$
    $I_q \leftarrow$ Reshape($[\langle\sigma_q^1\rangle, \langle\sigma_q^2\rangle, \cdots, \langle\sigma_q^m\rangle]^T$)
    $I_k \leftarrow$ Reshape($[\langle\sigma_k^1\rangle, \langle\sigma_k^2\rangle, \cdots, \langle\sigma_k^m\rangle]^T$)
    $I_v \leftarrow$ Reshape($[\langle\sigma_v^1\rangle, \langle\sigma_v^2\rangle, \cdots, \langle\sigma_v^m\rangle]^T$)
    Extract Text Features:
    Attention(I_q, I_k, I_v) $\leftarrow$ softmax($\frac{I_q {I_k}^T}{\sqrt{d_k}}$) I_v
    MultiHead(I_q, I_k, I_v) $\leftarrow$ Concat($head_1, head_2, \cdots, head_h$)$W_0$
    $Z' \leftarrow$ LayerNorm($H^{(0)}$+ MultiHead(I_q, I_k, I_v))
    FFN(Z') $\leftarrow$ ReLU($Z'W_1 + b_1$)$W_2 + b_2$
    Image_feature $\leftarrow$ LayerNorm(Z'+FFN(Z'))
End for
Return Image_feature

Algorithm 3. : Feature fusion

Data: T=Text_feature, I=Image_feature
Result: Prediction results
Initialization:
    Linear transformation:
    $T' \leftarrow TW_T + b_T$ and $I' \leftarrow IW_I + b_I$
    Get quary,key,value:
    $Q_T \leftarrow T'W_Q^T, K_T \leftarrow T'W_K^T, V_T \leftarrow T'W_V^T$
    $Q_I \leftarrow I'W_Q^I, K_I \leftarrow I'W_K^I, V_I \leftarrow I'W_V^I$
    Calculate attention score and weighting:
    $T_{fused} \leftarrow \text{softmax}(\frac{Q_T {K_I}^T}{\sqrt{d_k}})\, V_I$
    $I_{fused} \leftarrow \text{softmax}(\frac{Q_I {K_T}^T}{\sqrt{d_k}})\, V_T$
    Feature concat:
    $F^{(0)} \leftarrow \text{Concat}([T_{fused}, I_{fused}])$
    Through multiple layers of Transformer Encoder:
    For $i \in [1, N]$ do
        $F^{(i)} \leftarrow \text{TransformerEncoderLayer}(F^{(i-1)})$
    End for
    Obtain the final result through a classifier:
    $\text{Pred} \leftarrow \text{Classifier}(F^{(N)})$
Return Pred

CRediT authorship contribution statement

Zhang Jiahui: Resources, Project administration. **Li Daiyi:** Methodology, Investigation. **Han Yucan:** Software, Resources. **Kong Delong:** Writing – review & editing, Writing – original draft. **Wu Huaiguang:** Data curation, Conceptualization. **Wang Lijie:** Funding acquisition, Formal analysis.

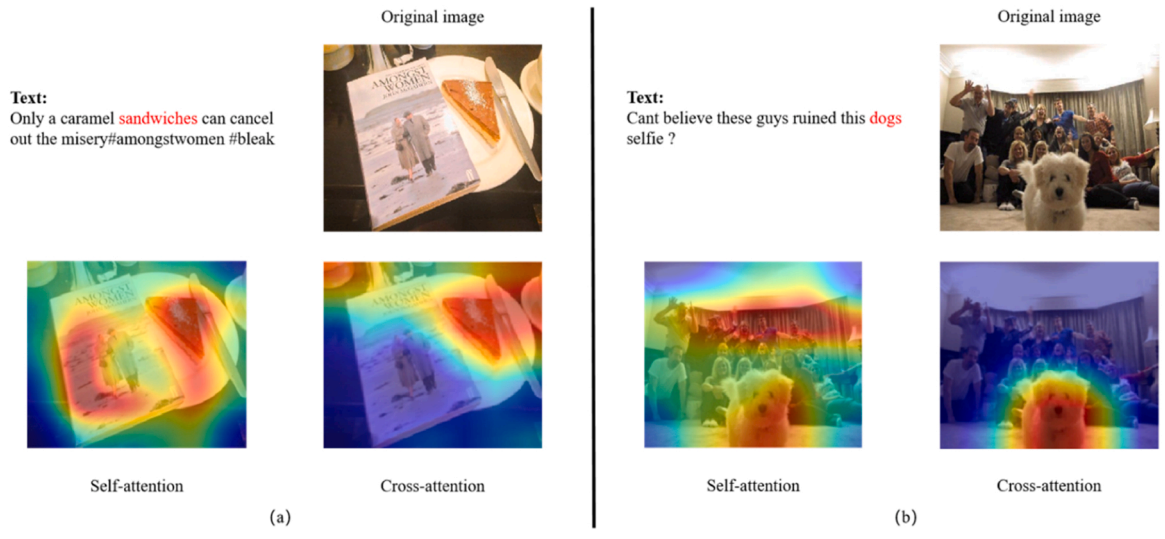


Fig. 9. Visualization results of cross attention and self-attention fusion module on image-text pairs selected from tweets.

Table 8
Comparison model test results with ITQT-MSA and CiteNet.

Images	Text	Real emotion	ITQT-MSA	CiteNet
	Enthusiastic hellofrom the kids ofZarqa.	Positive	Positive	Positive
	Even a packed Ed-inburgh to Glasgowevening train is funin the company ofebullient art writer,Anne Ellis.	Neutral	Neutral	Neutral
	CUT ofrece car-reras de cincoUniversidades	Positive	Positive	Positive
	Finished foundry.Now I just need safety gear, and I'll be popping out aluminum biscuits f…	Negative	Negative	Neutral

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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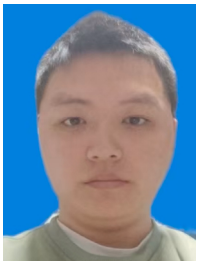
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